

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

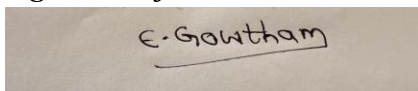
1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I Enamala Gowtham/192311190 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with the handwritten signature "E. Gowtham" in black ink. The signature is written in a cursive style with a horizontal line underneath the name.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

Q1) Fundamental Concepts in Web Development.

i) XML (Extensible Markup Language)

XML is a markup language used to store, structure, and transport data in a platform independent format. It allows developers to define their own tags according to the type of information being represented.

Example:

```
<student>
```

```
<name>Gowtham </name>
```

```
<regno>192311190 </regno>
```

```
<department>CSG </department>
```

```
</student>
```

ii) JSP (Java Server Pages)

JSP is a server-side java technology used to create dynamic web pages. It allows java code and web-related components to work together to generate HTML content dynamically.

JSP is commonly used to java-based web applications for processing user requests, accessing databases, and displaying dynamic

data. Jsp can also use technologies such as JSTL and EL to reduce the amount of java code placed directly in web pages.

iii) MVC Architecture

MVC stands for Model-View Controller. It is a software architectural pattern that separates an application into three components:

- Model
- View
- Controller

iv) XSLT (Extensible stylesheet language Transformations)

XSLT is a language used to transform XML documents into other formats, such as HTML, XML or text. It can select XML elements, rearrange data, and format information for permission.

v) XML Schema

XML schema, commonly called XSD, is used to define the structure, elements, attributes, and data types allowed in an XML document.

It can specify requirements such as:

- The order of elements
- minimum and maximum occurrences
- which elements must exist.

vi) JSTL (Jsp standard Tag Library)

JSTL is a collection of standard tags used in JSP applications. It provides features for iteration, conditional processing, formatting and working with data without requiring large amount of java code inside JSP pages.

vii) PHP CHypertext preprocessor)

PHP is a server-side scripting language widely used for developing dynamic web applications. PHP code is executed on the server, and the resulting HTML is sent to client browser.

viii) Database Connectivity

Database connectivity is the mechanism through which a web application communicates with data base to store, retrieve, update, and delete information.

Q2. Interconnection of the Technologies in a Web Application

The above technologies work together to create a data-driven web application. Each technology performs a specific function, which MVC provides an organized structure for integrating them.

i) XML as a Data Representation format.
XML represents data using a ~~her~~ hierarchical structure of elements and attributes.

For example:

```
<students>
  <student>
    <id>19231190</id>
    <marks>85</marks>
  </student>
</students>
```

ii) XML parsing

After XML is received or stored, an XML parser reads the documents. A DOM parser can represent the XML as a tree, allowing the application to access individual elements. A SAX parser processes the document sequentially.

iii) XSLT for XML Transformation

XSLT can transform XML data into a presentation-oriented format such as HTML

XML Data $\longrightarrow$ XSLT Transformation $\longrightarrow$ HTML

iv) JSP for server-side processing

JSP is used on the server to generate dynamic web content. In an MVC-based Java application, JSP generally acts as the view.

A JSP page can receive processed data from the controller and display it to the user. JSTL can be used within JSP to perform tasks such as iteration and conditional display while keeping presentation code cleaner.

v) PHP for server-side processing

PHP also performs server-side processing, but it belongs to a different server-side technology ecosystem from JSP. PHP can receive HTTP requests, process form data, perform business operations, access databases

user → Controller → Model → Database/XML
→ Controller → View → user

Overall Integration

client request



Controller (MVC)



Model / Business Logic



Database Connectivity / XML



XSLT Transformation



JSP/JSTL view



HTML Response to client